



End Term (Even) Semester Examination May-June 2025

Roll no.....

Name of the Program and semester: MCA 4th Semester

Name of the Course: Advanced Graphics and Visual Computing

Course Code: TMC 403(2)

Time: 3 hour

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

- Q1. (2X10=20 Marks)
- a. Implement the DDA line Algorithm in C Programming. [CO1]
 - b. Translate an object defined by A (0, 0), B (1, 0), C (1, 1) and D (0, 1) by 3 unit in X and Y direction respectively. [CO1]
 - c. Differentiate 3D Translation and 3D Scaling with the help of matrix representations. [CO1]
- Q2. (2X10=20 Marks)
- a. Design and implement the Flood-Fill Algorithm in C Programming. [CO2]
 - b. Define line clipping? Write down the program in C for Cohen Sutherland Algorithm. [CO2]
 - c. Differentiate orthographic & oblique projection. [CO2]
- Q3. (2X10=20 Marks)
- a. Draw Spline representations for a curve. [CO3]
 - b. Differentiate Bezier Curves & surfaces with example. [CO3]
 - c. Demonstrate Rational splines. [CO3]
- Q4. (2X10=20 Marks)
- a. Design a program for Painter's algorithm. [CO4]
 - b. Differentiate Affine and Coordinate system transformations. [CO4]
 - c. Explain Basic Rendering techniques. [CO4]
- Q5. (2X10=20 Marks)
- a. Differentiate Isosurfaces and Isocontours and explain how to create 2D Isocontours with Matplotlib. [CO5]
 - b. Differentiate Visualization of 2D/3D scalar fields. [CO5]
 - c. Explain Effective Color Mapping by Code in Python (Using Matplotlib). [CO5]